

Dissolved Oxygen at Liquid Boundaries

Liquid Surface

Boundary condition at liquid surface:

$$-D_{O_2,l} \frac{dC_{O_2,l}}{dz} \Big|_{\text{surface}} = k_{O_2} \left(\overline{C_{O_2,g}} - \frac{C_{O_2,l} \Big|_{\text{surface}}}{S_F \cdot \beta_{O_2} \cdot R \cdot T_l} \right)$$

Using finite differences in space and expanding:

$$\begin{aligned} -D_{O_2,l} \frac{(C_{O_2,l} \Big|_{\text{surface}+\Delta z} - C_{O_2,l} \Big|_{\text{surface}})}{\Delta z} &= k_{O_2} \cdot \overline{C_{O_2,g}} - \frac{k_{O_2} \cdot C_{O_2,l} \Big|_{\text{surface}}}{S_F \cdot \beta_{O_2} \cdot R \cdot T_l} \\ C_{O_2,l} \Big|_{\text{surface}} \cdot \left(\frac{1}{\Delta z} + \frac{k_{O_2}}{D_{O_2} \cdot S_F \cdot \beta_{O_2} \cdot R \cdot T_l} \right) &= \frac{k_{O_2} \cdot \overline{C_{O_2,g}}}{D_{O_2}} + \frac{C_{O_2,l} \Big|_{\text{surface}+\Delta z}}{\Delta z} \\ C_{O_2,l} \Big|_{\text{surface}} &= \frac{\left(\frac{k_{O_2} \cdot \overline{C_{O_2,g}}}{D_{O_2}} + \frac{C_{O_2,l} \Big|_{\text{surface}+\Delta z}}{\Delta z} \right)}{\left(\frac{1}{\Delta z} + \frac{k_{O_2}}{D_{O_2} \cdot S_F \cdot \beta_{O_2} \cdot R \cdot T_l} \right)} \end{aligned}$$

Liquid Bottom

Boundary condition at liquid bottom (with cells):

$$\begin{aligned} -D_{O_2,l} \frac{dC_{O_2,l}}{dz} &= \frac{MR_{O_2}}{A_s} \\ MR_{O_2} &= \frac{R_c \cdot N_c \cdot C_{O_2,l} \Big|_{\text{bottom}}}{C_{O_2,l} \Big|_{\text{bottom}} + k_c} \end{aligned}$$

Using finite differences in space and substituting:

$$\begin{aligned} -D_{O_2,l} \frac{(C_{O_2,l} \Big|_{\text{bottom}} - C_{O_2,l} \Big|_{\text{bottom}-\Delta z})}{\Delta z} &= \frac{R_c \cdot N_c \cdot C_{O_2,l} \Big|_{\text{bottom}}}{A_s (C_{O_2,l} \Big|_{\text{bottom}} + k_c)} \\ -D_{O_2,l} \cdot (C_{O_2,l} \Big|_{\text{bottom}} - C_{O_2,l} \Big|_{\text{bottom}-\Delta z}) \cdot (A_s \cdot C_{O_2,l} \Big|_{\text{bottom}} + A_s \cdot k_c) &= \Delta z \cdot R_c \cdot N_c \cdot C_{O_2,l} \Big|_{\text{bottom}} \\ -D_{O_2,l} \cdot A_s \cdot C_{O_2,l} \Big|_{\text{bottom}}^2 - D_{O_2,l} \cdot A_s \cdot k_c \cdot C_{O_2,l} \Big|_{\text{bottom}} + D_{O_2,l} \cdot A_s \cdot C_{O_2,l} \Big|_{\text{bottom}} \cdot C_{O_2,l} \Big|_{\text{bottom}-\Delta z} &+ D_{O_2,l} \cdot A_s \cdot k_c \cdot C_{O_2,l} \Big|_{\text{bottom}-\Delta z} = \Delta z \cdot R_c \cdot N_c \cdot C_{O_2,l} \Big|_{\text{bottom}} \end{aligned}$$

Dividing by $-D_{O_2} \cdot A_s$ and combining terms:

$$C_{O_2,l} \Big|_{\text{bottom}}^2 + \left(k_c - C_{O_2,l} \Big|_{\text{bottom}-\Delta z} + \frac{\Delta z \cdot N_c \cdot R_c}{D_{O_2} \cdot A_s} \right) \cdot C_{O_2,l} \Big|_{\text{bottom}} - k_c \cdot C_{O_2,l} \Big|_{\text{bottom}-\Delta z} = 0$$

Use quadratic equation (positive root to avoid negative concentrations) to determine $C_{O_2,l}\Big|_{\text{bottom}}$

Coefficients: $\mathbf{a} = 1$, $\mathbf{b} = k_e - C_{O_2,l}\Big|_{\text{bottom}-\Delta z} + \frac{\Delta z \cdot N_c \cdot R_c}{D_{O_2} \cdot A_s}$, $\mathbf{c} = -k_e \cdot C_{O_2,l}\Big|_{\text{bottom}-\Delta z}$

$$C_{O_2,l}\Big|_{\text{bottom}} = \frac{-\left(k_e - C_{O_2,l}\Big|_{\text{bottom}-\Delta z} + \frac{\Delta z \cdot N_c \cdot R_c}{D_{O_2} \cdot A_s}\right)}{2} + \frac{\sqrt{\left(k_e - C_{O_2,l}\Big|_{\text{bottom}-\Delta z} + \frac{\Delta z \cdot N_c \cdot R_c}{D_{O_2} \cdot A_s}\right)^2 + 4 \cdot k_e \cdot C_{O_2,l}\Big|_{\text{bottom}-\Delta z}}}{2}$$